

Labwork on “Ensemble learning”

Advanced machine learning
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Solutions will be given *here*

Exercise 1: feature importance (result interpretation)

- (a) Load the 'diabetes' dataset (from sklearn.datasets).
- (b) Validate the predictive power of a random forest and a xgb regressor both with 500 trees on the diabetes dataset.
- (c) Train the previous models on the whole dataset and evaluate them on the same data.
- (d) Compute 3 types of feature importance according to the 2 regressors; Gini, permutation and Shapley-based ones, plot them and conclude.

Exercise 2: models comparison

- (a) Load the 'wine' dataset (from sklearn.datasets) and plot the corresponding correlation matrix with a clear and readable heatmap that respect the following constraints:
 - the legend should include extreme correlations values (i.e. -1 and +1)
 - negative/positive correlations should be represented respectively by cold/warm colours
 - x-labels have to be rotated of 45°
 - all variable names have to be clearly readable

- (b) With the *sklearn* package set up a model comparison pipeline between a *dummy* classifier (predicting systematically the most frequent label of the training data), a decision tree, a kNN and a neural network models with a random sample containing 3/4 of the data for training and 1/4 for testing. Repeat this operation 100 times, draw the results in terms of means and boxplots and conclude.

N.B.: Unless specified in the question, the solver to be used for neural network is 'adam', the trees depths are fixed to 1, kNN models are tuned to 2 neighbors and neural network are limited to MLP of 1 hidden layer of 10 neurons computed with a maximum of 1000 iterations.

- (c) Add hard and a soft votes approaches to the pipeline of question (b) and comment the results.
- (d) Try a stacking approach (with the same 3 single classifiers as before), add it to the comparison of the previous question and comment the results.
- (e) Add a random forest, an AdaBoost, a gradient boosting and a XGBoost to the pipeline and comment the results.

- (f) Tune the forest size (i.e. number of trees) with a grid-search approach and with a Bayes-search approach on the whole dataset.
- (g) Realise a comparison between random forest and XGB with a 10-fold cross validation procedure after having tuned their sizes (number of weak classifiers) and their depths on the whole dataset. What is wrong with our tuning strategy?
- (h) Realise the complete models comparison of previous question with a proper tuning step for all models' main hyperparameters.

NB: Due to the long computation times involved in that question, no results are expected, only the codes have to be written.

Exercise 3: order of ensemble learning

Should I train 1 forest of 10 trees, 2 forests of 5 trees or 5 forest of 2 trees?

- (a) Load the 'BreastCancer' dataset (from sklearn.datasets on python or from the mlbench package on R).
- (b) Answer the main question with a model comparison (involving simple vote aggregations) and plot the results.
- (c) Do the same thing replacing the forest approach with stacking and boosting approaches.